

# HW 3

## 1

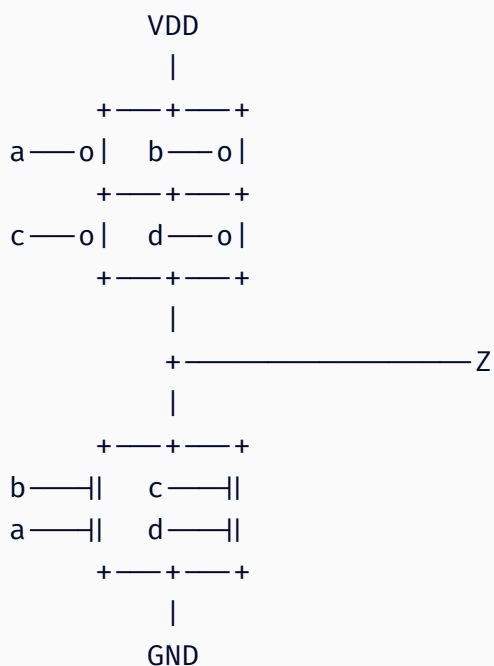
For the AND-OR-INVERT (AOI22) circuit derive:

$$f = (ab + cd)'$$

### a

Complex CMOS gate

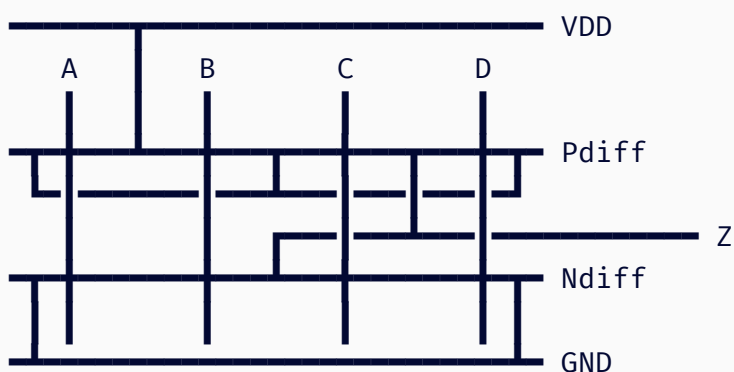
✓ Answer ✓



### b

corresponding stick diagram.

✓ Answer





Compare the two networks

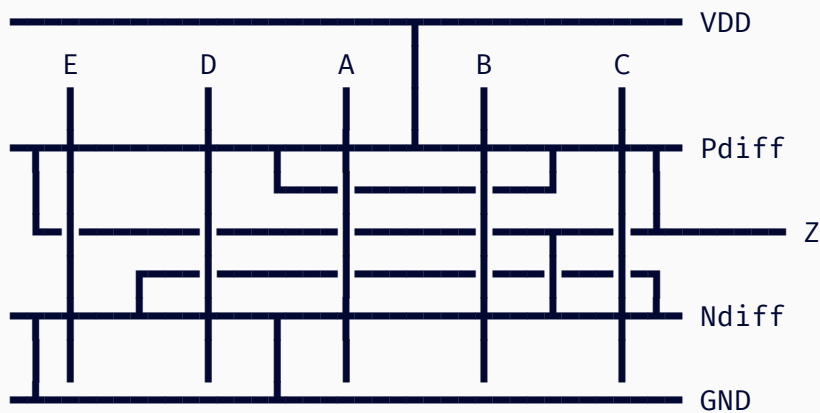
✓ **Answer**

The n-network has a bridge optimization that is not possible on the p-network.

3

Complete the diagram in (b) so that it implements the circuit in (a).

✓ **Answer**

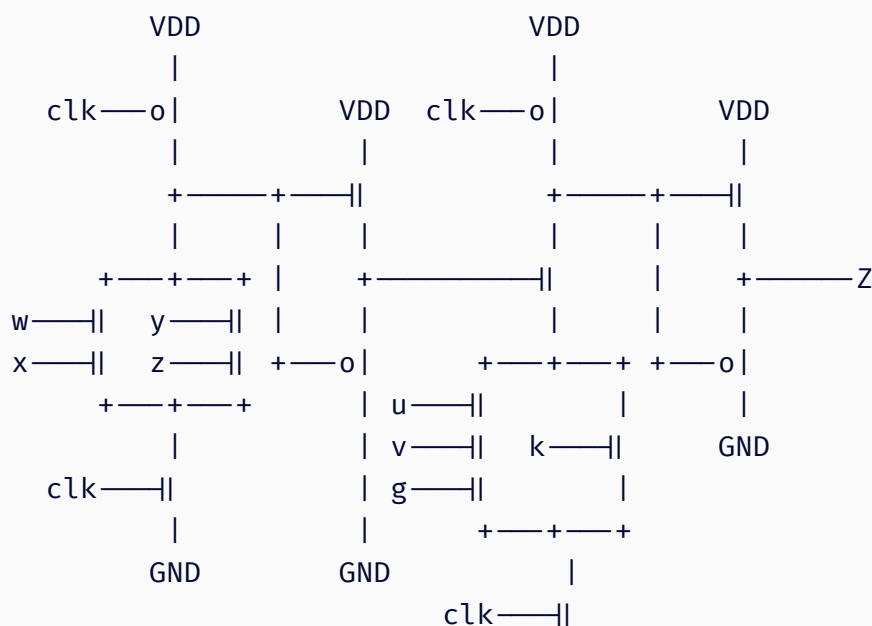


4

a

Design a CMOS logic gate for the function  $Z = (wx + yz)(uv + k)$  using N-Domino logic (Do not connect more than 4 transistors in series; Assume single rail logic).

✓ **Answer**

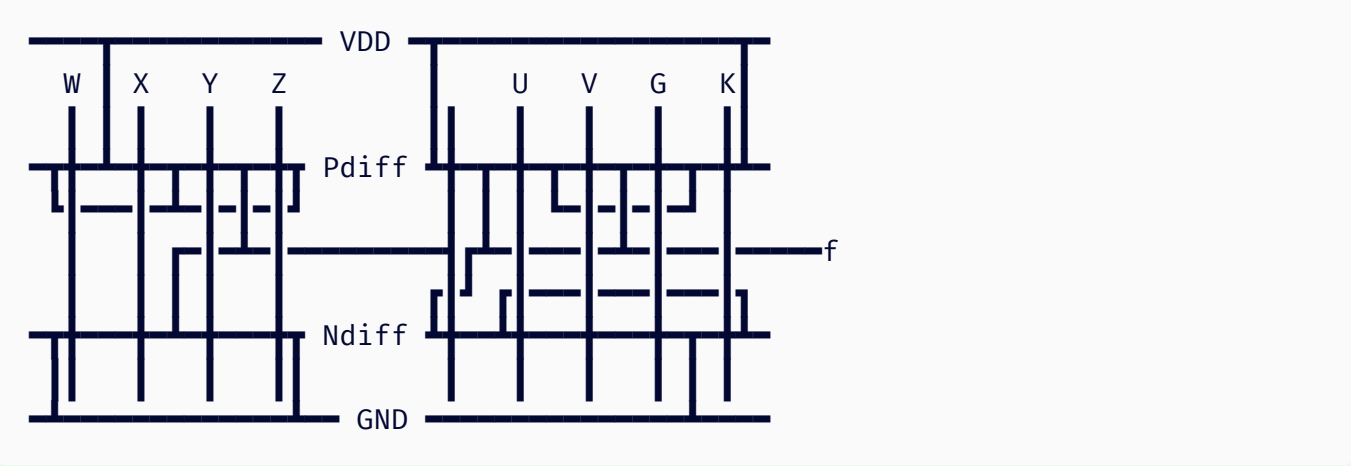


I  
GND

b

Derive a stick diagram for your CMOS gate.

✓ Answer



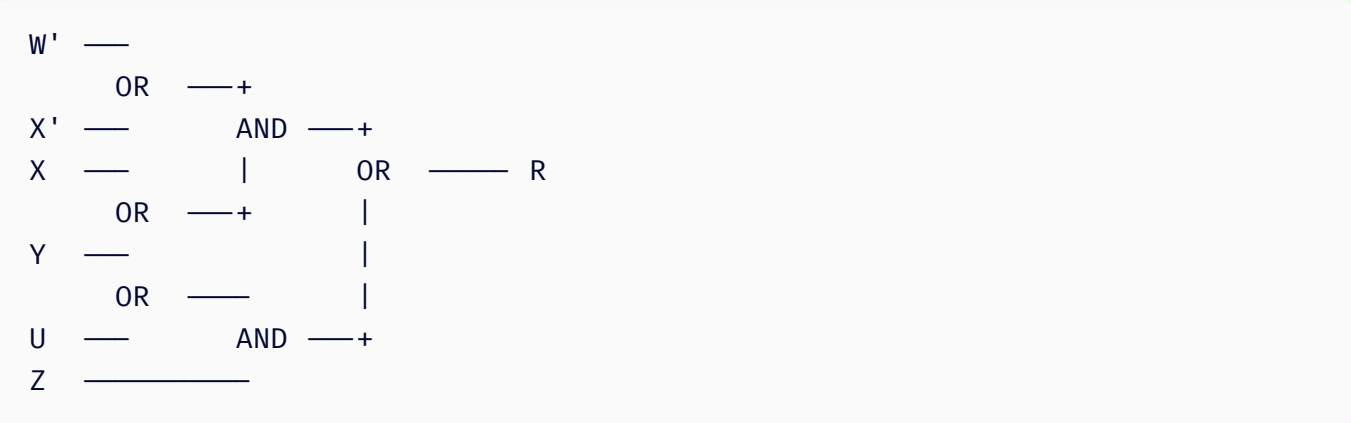
5

Assume that both inverted signals is available for every primary input

a

Convert the gate design into one that lends itself to a domino CMOS implementation. Do not write out logic equations, minimize gates or merge signals to create complex gates without rewiring, but merely by changing the existing gates and adding and removing inverters, you are to create 5 domino gates that compute the same value for  $Z$  as the circuit above. Use De Morgan's Law to transform existing gates where necessary. The inputs  $(w, x, y, u, z)$  are driven by domino gates and the output,  $R$ , must be able to drive a domino gate (use only two basic DOMINO CMOS cells);

✓ Answer



b

Derive a CMOS implementation for the logic gate circuit using only three basic CMOS cells

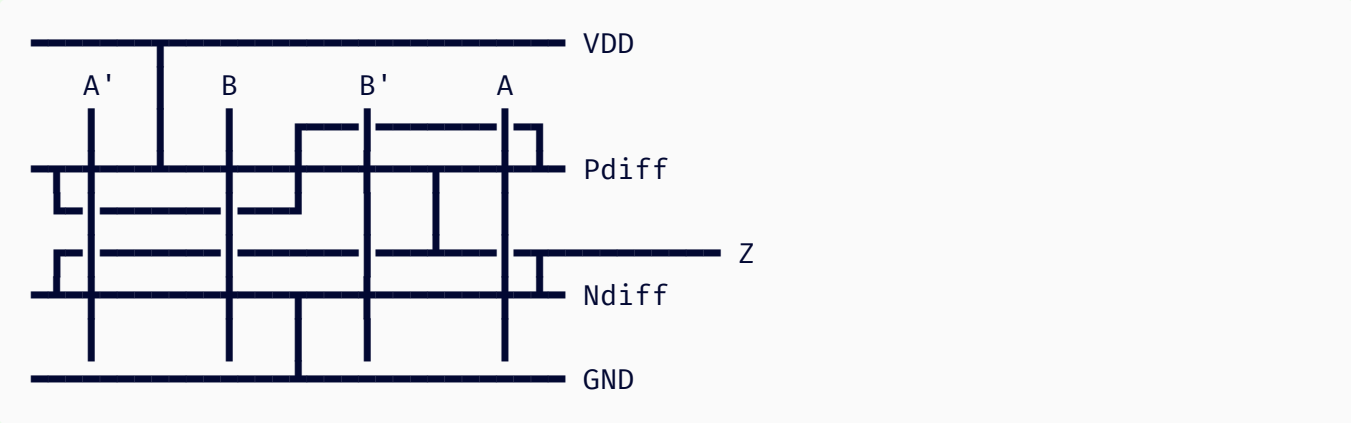
✓ Answer

OR, AND, and NAND

6

Complete the stick diagram so that it implements the circuit shown below (You may use two layers for metal)

✓ Answer



7

Label the inputs of the circuit below so that it implements the function  $Z = d(b + b'c) + abc' + a'bc$

✓ Answer

From top to bottom, A, C, B, D

8

Draw a Stick's Diagram for the master stage of a D flip-flop shown below. Vdd and Vss should run horizontally in metal1 at the top and bottom of the cell, respectively, D should enter from the left side in metal, Q should exit from the right side in metal1, and  $\phi$  and  $\phi'$  should run vertically in Polysilicon (a single line each).

✓ Answer

